2023_Hexaware_Actual_Test_8

Test Summary

- No. of Sections: 2
- No. of Questions: 90
- Total Duration: 90 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. In a boarding house, the number of boarders increases at a certain rate per annum. Four years ago the number of boarders was 49, now it is 196. What will be two years hence?

- 196
- 392
- 588
- 391

Q2. Hansie made the following amounts in seven games of cricket in India: Rs.10, Rs.15, Rs.21, Rs.12, Rs.18, Rs.19 and Rs.17 (all figures in crores of course). Find his average earnings.

- Rs. 16 crores
- Rs. 11 crores
- Rs. 17 crores
- Rs. 15.5 crores

Q3. A question paper has two parts P and Q, each containing 10 questions. If a student needs to choose 8 from part P and 4 from part Q, in how many ways can he do that?

- None of the mentioned options
- 6020
- 1200
- 9450

Q4. A bag contains nine distinct yellow balls, three distinct white balls, and four distinct red balls. In how many ways can two balls be drawn from the bag?

- $9C_2$
- $3C_2$
- $16C_2$
- $12C_2$

Q5. What is the probability of getting a sum 9 from two throws of a dice?

- 5/9
- 8/9
- 1/9
- 2/9

Q6. A bag contains 3 green, 5 black and 4 brown balls.3 balls are drawn randomly. What is the probability that all balls are different ?

- 11/3
- 2/11
- 11/2
- 3/11

Q7. 4 technicians and 6 trainees can diagnose a machine in 8 hours, while 3 technicians and 7 trainees can diagnose it in 10 hours. In how many hours will 10 trainees diagnose it?

- 32 Days
- 34 Days
- 36 Days
- 40 Days

Q8. 11 men working for 18 hours a day can complete a piece of work in 25 days, in how many days can 18 men working for 5 hours a day complete the same piece of work?

50 days

48 days

52 day

55 days

Q9. A train 150 m long, passes a pole in 15 seconds. And crosses the second train of same length travelling in opposite direction in 12 seconds. The speed of the second train is

48 kmph

40 kmph

52 kmph

54 kmph

Q10. A goods carriage of length 2 km headed to Srinagar from Punjab was running at a speed 30 km/hr. It crosses a tunnel which is 58 km long with that speed. Find the time taken by the goods carriage to cross the tunnel

4 hrs

2 hrs

3 hrs

1 hrs

Q11. 1 kg of tea and 4 kg of sugar cost Rs.88.20, but if sugar cost 50 percent and tea 10 percent, these would cost Rs.114.30. Find the price per kg of sugar:

Rs 10.8

Rs 45

Rs 15

Rs 11.25

Q12. Two mixed schools have 90 and 120 children respectively. In the first 60% and in the second 50% of the children are boys. What % of the children in the two schools are boys ?

54%

42%

65%

38%

Q13. Visvesh makes a profit of 22% on selling article A and makes a loss of 8% on selling article B, which overall gives him no profit/loss at the end. If he bought both the articles for a total amount of Rs. 600, then find the SP of article B?

Rs.404.80

Rs.440

Rs.536.80

Rs.160

Q14. In a certain store, the profit is 300% of the cost. If the cost increases by 60% but the selling price remains constant, approximately what percentage of the selling price is the profit?

70%

68%

60%

45%

Q15. A sum of Rs.10,000 deposited at compound interest becomes double after 5 years, how much will it become after 20 years?

Rs.160000

Rs.100000

Rs.50000

Rs.120000

Q16. Rs. 12000 amounts to Rs. 20736 in 3 years at r% p.a. of compound interest. What is the value of r?

10%

25%

12%

20%

Q17. The average weight of 7 women is increased by 12 kg when one of them whose weight is 95 kg is replaced by another woman. What is the weight of the new women (in kg)?

179

149

159

189

Q18. In a 26-question test, 5 points were deducted for each wrong answer, and 8 points were added for right answers. If all the questions were answered, how many were correct if the score was zero

10

11

12

13

Q19. If 2x = 3y = 4z, then what is the value of x : y : z?

1 : 3 : 5

1 : 4 : 5

8 : 3 : 6

6 : 4 : 3

Q20. In what ratio should a 30% base solution be mixed with a 50% base solution so that the resultant solution has a 40% base solution in it?

2:1

1:1

1:2

3:1

Q21. **Directions:**
What will come in place of question mark (?) in the following series?

582, 751, 555, 724, 528,?.

Q22. **Directions: What will come in place of question mark (?) in the following series?**

43, 47, 56, 81, 130, ?

Q23. A lady pointing to a man in a photograph says, "The sister of the son of this man is my mother-in-law." How is the husband of the lady related to the man in the photograph?

Father

Son

Maternal Grandson

Maternal Grandfather

Q24. Pointing to a gentleman, Abhishek said, “His only brother is the father of my daughter’s father. “How is the gentleman related to Abhishek?

Grand father

Father

Brother-in-law

Uncle

Q25. **Directions:**
In the following question, only one Conclusion is given and five statements are given as 1), 2), 3), 4) and 5). From this, you have to take the statements to be true even if they seem to be at variance with commonly known facts and then decide which of the given statement logically follows.

Conclusions:
All dogs being monkeys is a possibility.
Some Monkeys are Cats is a possibility

Statements:
Statements – 1: All dogs are cats. No cat is Pig. All Pigs are Monkeys.
Statements – 2: No Cat is dog. No dog is Monkey. No Monkey is pig.
Statements – 3: All dogs are cats. No dog is Pig. No cat is monkey.
Statements – 4: All Cats are dogs. No cat is Pig. No dog is Monkey.
Statements – 5: All dogs are cats. No cat is Monkey. All Pigs are Monkeys.

Only Statements – 1

Only Statements – 2

Only Statements – 3

Only Statements – 4

Q26. **Given two statements, verify the conclusions and mark the answer**

Statements:
Some cats are dogs.
No dog is cow.

Conclusions:
I. No cow is cat.
II. No dog is cat.

if only conclusion I follows.

if only conclusion II follows.

if both conclusions I & II follow.

if no conclusion follows.

Q27. **The question below consists of a question and two statements numbered (I) and (II) given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both statements and answer.**

What is the code for 'or' in the code language?
I) 'nik sa te' means 'right or wrong', 'ro da nik' means 'he is right', and 'fe te ro' means 'that is wrong'.
II) 'pa nik la' means 'that right man', 'sa ne pa' means 'this or that', and 'ne ka re' means 'tell this there'.

I alone is sufficient while II alone is not sufficient

II alone is sufficient while I alone is not sufficient

Either I or II is sufficient

Neither I nor II is sufficient

Both I and II are sufficient

Q28. **The question below consists of a question and two statements numbered (I) and (II) given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both statements and answer.**

On which date in August was Kapil born?
I) Kapil's mother remembers that Kapil was born before the nineteenth but after the fifteenth.
II) Kapil's brother remembers that Kapil was born before the seventeenth but after the twelfth.

I alone is sufficient while II alone is not sufficient

II alone is sufficient while I alone is not sufficient

Either I or II is sufficient

Neither I nor II is sufficient

Both I and II are sufficient

Q29. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions.**

Statement: I ≤ S, S < X, X = F, F ≤ K

Conclusion:
I. K > I
II. I < X

if only conclusion 1 is true

if only conclusion 2 is true

if either conclusion 1 or conclusion 2 is true

if neither conclusion 1 nor conclusion 2 is true

if both conclusion 1 and conclusion 2 are true

Q30. **In these questions, a relationship between elements is shown in the statements. The statements are followed by two conclusions. Give answer.**

Statements: U = V ≤ W > Q ≥ N; B < Q; E = W

Conclusion: I. E > Q II. W > B

If only conclusion I is true

If only conclusion II is true.

If either conclusion I or II is true.

If neither conclusion I nor II is true.

If both conclusion I and II are true.

Q31. If FRIEND is coded as HUMJTK, how is CANDLE written in that code?

EDRIRL

DCQHQK

ESJFME

DEQJQM

Q32. In the following question, the alphabets bear a Random Sequence Numbering. Find the Numerals that would be represented by the following words.

If ANTI is coded as 7 -20 – 26 -15, then how would you code POST?

22 -21 -25 -26

21 -22 -23 -24

26 -25 -24 -23

19 -21 -25 -26

Q33. **Find the next term in the following series.**

QAR, RAS, SAT, TAU, _____

UAV

UAT

TAS

TAT

Q34. Choose the correct order of the given alternative :
m n o n o p q o p q r s _ _ _ _ _

mnopq

oqrst

pqrst

qrstu

Q35. Two statements numbered I and II are given below. You have to decide whether the data provided in the statements are sufficient to answer the question.

What is the code for 'juicy apple'?

Statements:
I. The code for 'apple juice is cedar' is 'ka la do ni', the code for 'cedar vinegar is good' is 'do re la mi' and the code for 'buy some juicy apples' is 'vu hi wo pe'.
II. The code for 'good orange and apple' is 'ni xe co mi', the code for 'apple juice is fresh' is 'ka la ni po' and the code for 'fresh and juicy orange' is 'xe co pe po'.

If the data in statement I alone are sufficient to answer the question while data in statement II alone is not sufficient.

If the data in statement II alone is sufficient to answer the question while the data in statement I alone is not sufficient.

If the data in both statements I and II together are not sufficient to answer the question

If the data either in statement I alone or in statement II alone is sufficient to answer the question.

If the data in both statements I and II together are necessary to answer the question

Q36. **Two statements numbered I and II are given below. You have to decide whether the data provided in the statements are sufficient to answer the question.**

In which direction is Kamal facing when he reached point Q?

I. Kamal starts from point M and walks 10m to his east and reaches N where he takes a right and walks 15m to reach point O.
II. Kamal moves towards North from point O and walks 50m to reach point P and finally takes a left turn and walks 20m to reach point Q.

If the data in statement I alone are sufficient to answer the question while data in statement II alone is not sufficient.

If the data in statement II alone is sufficient to answer the question while the data in statement I alone is not sufficient.

If the data in both statements I and II together are not sufficient to answer the question

If the data either in statement I alone or in statement II alone is sufficient to answer the question.

If the data in both statements I and II together are necessary to answer the question

Q37. **Find the odd man out:**

2D, 6E, 14J, 25P

25P

6E

2D

14J

Q38. **Complete the following sentences by choosing the right words for the blanks given.**

A highly intelligent person often thinks (i)____; a few snippets of information can trigger a (ii)____ conclusion that might not stand up to closer and (iii)____ scrutiny.

Blank (i)
A. too deeply
B. too warily
C. too quickly

Blank (ii)
D. firm
E. labored
F. hasty

Blank (iii)
G. slower
H. precipitous
I. overt

CFG

BEI

CEG

AEG

Q39. **Select the word from the given options that can fit the blanks in all the sentences given in the questions.**

A. Please print your name and address in _____ letters.
B. My friend and I live on the same _____
C. I'm not going to put my head on the _____ for you.
D. Every autumn dead leaves _____ the drains.

capital

block

stop

range

Q40. **Pick out the best alternative for the underlined part in the given sentences**

She continued her singing, dancing and acting in plays since her school and college years but lately get more and more involved in social work.

since her school and college years but lately get

since her school and college years but lately got

throughout her school and college years but lately got

for her school and college years but lately get

Q41. **Which of the phrases (1), (2), (3) and (4) given below each sentence should replace the underlined phrase to make the sentence grammatically meaningful and correct?**

The thief was taken away when he saw the policeman and silently slipped out of that place.

had been taken away

was taken aback

will be taken aback

would be taken away

Q42. **Pick out the best alternative for the underlined part of sentence.**

State control has always and will continue to be feared until socialism will have taken over.

is always feared and until socialism takes over, will continue to be so

is and will always continue to be feared until socialism takes over

has always been and will continue to be feared until socialism takes over

continues being feared and will be feared by people until the takeover by socialism

Q43. **Replace the underlined part of the sentence with a grammatically correct option.**

Does he thought about the consequences if he refuses the offer of someone so powerful?

Does he thought about

Does he know

He will have to face

Had he thought about

Q44. **Substitute the given sentences with one word.**

Everyone in the meeting praised Rama for his presentation, suddenly Rakesh from behind stood up **to raise his opposition.** Rakesh is ____.

- Protagonist
- Sidekick
- Antagonist
- Comrade

Q45. Find the opposite of :
Subvert

- Corrupt
- Profane
- Combat
- Elevate

Q46. Pick out the word most similar in meaning.

Abeyance

- Temporary suspension
- Abstinence
- Hatred
- Memory

Q47. **Convert the voice of the given statement:**

Rohit was taken to the hospital by the villagers.

- Rohit was helped by the villagers to reach the hospital
- The villagers took Rohit to the hospital
- The villagers has reached the hospital with Rohit

The villagers reached the hospital with Rohit

Q48. **Convert the below sentence into its active/passive form:**

By now the participants will have been vacated.

They will vacated the participants now.

They will have vacated the participants by now.

They have vacated the participants by now.

They will have been vacated the participants by now.

Q49. **In each of the following questions, four statements A, B, C and D are provided between an opening statement 1 and a closing statement 6. The four statements are jumbled up and form a coherent paragraph when properly arranged between the two given statements. Select the alternative representing the proper and logical sequencing of these six taken together.**

1. At the heart of the novel – as dark as Conrad's – is the relationship between Piya and her guide Fokir.
(A) He can read the secrets she is searching for; he, the child of the river, can see them before her binoculars reach out to them.
(B) In his ramshackle boat unfolds the best of this novel.
(C) They have no words in common.
(D) The river unites them; it reduces the cultural – and linguistic – gap between them.
6. Silence and gestures, movements of eyes and figures, translate their emotions for each other and dramatize the tension, elemental, erotic.

BADC

BDAC

DBAC

DABC

Common Content:

Study the following table carefully and answer the questions given below.

Amount invested by six different companies during six different months (in lakhs).

Company Months	A	B	C	D	E	F
January	25	28	50	32	24	35
February	45	40	60	55	28	38
March	52	38	46	72	83	65
April	80	45	70	90	65	76
May	15	20	47	25	13	30
June	62	95	27	80	50	20

Q50. What is the average investment made by company C in May, D in February and E in January (in lakhs)?

22

36

42

46

Q51. What is the difference between the investment made by company A and E together in March and company B and D together in January (in lakh)?

75

65

50

85

Q52. By what percent approximately is the investment made by company A in June less than that by Company D in the same month?

23%

12%

35%

42%

Common Content:

Read the following passage carefully and answer the respective questions given below each of them.

What are the good parts of our civilization? First and foremost there are order and safety. If today I have a quarrel with another man, I do not get beaten merely because I am physically weaker and he can kick me down. I go to law, and the law will decide as fairly as it can between the two of us. Thus in disputes between man and man right has taken the place of might. Moreover, the law protects me from robbery and violence. Nobody may come and break into my house, steal my goods, or run off with my children. Of course, there are burglars, but they are very rare, and the law punishes them whenever it catches them. It is difficult for us to realize how much this safety means. Without safety, these higher activities of mankind which make up civilization could not go on. The inventor could not invent, the scientist found out or the artist made beautiful things. Hence, order and safety, although they are not themselves civilization are things without which civilization would be impossible. They are as necessary to our civilization as the air we breathe is to us, and we have grown so used to them that we do not notice them any more than we notice the air.

Another great achievement of our civilization is that today civilized men are largely free from the fear of pain. They still fall ill, but illness is no longer the terrible thing it used to be... Not only do men and women enjoy better health; they live longer than they ever did before, and they have a much better chance of growing up. Thirdly, our civilization is more secure than any that have gone before it. This is because it is much more widely spread... Previous civilizations were specialized and limited, they were like oases in a desert.

Q53. In olden days the rule of life was :

Right

- Might
- Spiritualism
- Money
- None of these

Q54. What according to the author has the freedom from the fear of pair led to?

- Better health
- Devotion to duty
- Increased cultural pursuits
- Abundant Leisure

Q55. What is the first merit of our civilization?

- Material advancement
- Cultural advancement
- Development of science
- Order and safety
- Spiritualism has become a way of life

Q56. The third merit of the present civilization, according to the author, is

- The present civilization is founded on justice
- Thepresent civilization has liberal outlook on life
- The present civilization has provided more leisure
- The present civilization is more wide spread in the sense that large number of people are civilized now than ever before
- The present civilization is more secure than any that has gone before

Q57. What according to the author, is the second merit of the present civilization?

- Development of means of transport and communication

Freedom from drudgery

Freedom from the fear of pain

Mental enlightenment of the people

Common Content:

Fill in the blanks using the options given below:

Cloud computing is a method for delivering information technology (IT) services in which resources are _____1_____ from the Internet through web-based tools and applications, as opposed to a direct connection to a server. Rather than keeping files on a _____2_____hard drive or local storage device, cloud-based storage makes it possible to save them to a remote database. The rise of cloud-based software has offered companies from all sectors several benefits, including the ability to use software from any device, either via a native app or a browser. As a result, users can carry their files and settings to other devices in a completely _____3____manner. Cloud computing offers big businesses some serious cost-saving potential. Before the cloud became a ___4____alternative, companies were required to purchase, construct and maintain costly information management technology and infrastructure. Now, instead of investing millions in huge server centers, a firm can use "lite" versions of workstations with _____5_____ fast internet connections.

Q58. Choose the right word for the blank numbered (3)

Viable

Seamless

Retrieved

Lightening

Q59. Choose the right word for the blank numbered (5)

Retrieved

Seamless

Lightening

Viable

Q60. Choose the right word for the blank numbered (4)

Viable

Tangible

Lightening

Seamless

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode?

```
1  integer i, j
2  set i=3, j=5
3  while(i>0)    //line
4  i=i-1
5  while(j<6)
6  print "Hello"
7  j=j+1
8  Go to line
9  end while
10 end while
```

- It will print Hello two times
- It will print Hello one time
- It will print Hello four times
- It will print Hello three times

Q2. What will be the output of the following pseudocode?

```
1 Integer p,q,r
2 Set p=4, q=5,r=6
3 q=3+r
4 p=q&p
5 r=q+r
6 p=p+q
7 Print p+q+r
```

- 38
- 33
- 19
- 53

Q3. What will be the output of the following pseudocode?

```
1 Integer b=7, c=1, i, j
2 for (j=2 to 3)
3    for (i = 1 to J)
4       c = c*2
5    end loop
6 end loop
7 b= b+c
8 Print b, c
```

36 43

23 16

71 64

39 32

Q4. What will be the output of the following pseudocode?

```
1  Integer a, b, c
2  set a=5, b= 11, c= 7
3  for ( each c from 2 to 3)
4  b = 9 + a
5  end for
6  print a + b
```

24

20

15

19

Q5. What will be the output of the following pseudocode?

```
1  Integer a, b, c
2  Set a=0, b=6, c=9
3  for(each c from 2 to 5)
4    a = 4 + c
5    if((c+b) <(b-c))
6      b=(b^a)+b
7      a= 10&a
8    End if
9  End for
10 Print a+b
```

24

11

19

15

Q6. What will be the output of the following pseudocode if **a=2**?

```
1  Integer fun(integer a)
2    if(a>0)
3      return a+ fun(a-1)
4    else
5      return 0
6    end if
```

```
7  End Integer fun()
```

- 6
- 5
- 3
- 4

Q7. What will be the output of the following pseudocode for **x=75**?

```
1  Integer fun(integer x)
2    if((x - 35)>(x /4))
3       return (x/2)+5
4    else
5       return fun(fun(x/2))
6    end if
7  End function fun ()
```

- 44
- 42
- 22
- 28

Q8. What will be the output of the following pseudocode?

```
1   integer a, b, c
2   set b= 300, a= 5, c= 1
3   if( b> a)
4   b= a
5   else
6   a= b
7    end if
8   for ( each b from 0 to 5)
9   a= a+ 1
10  end for
11  print a + c
```

- 15
- 13
- 1
- 12

Q9. What will the output of the following pseudocode for b = 18?

```
1  Integer calculate(integer b)
```

```
2  If (b EQUALS 1)
3  Return 0
4  Else
5  Return 5 + calculate(b/2)
6  End function calculate ()
```

- 34
- 20
- 21
- 56

Q10. What will be the output of the following pseudocode?

```
1  integer p, q, r
2  set  p=1, q=4, r=2
3  if (p ^q  > p&q )
4  r= p& q
5  else
6  r=p^ q
7  end if
8  if(q> r  &&  r>p)
9  p = q
10 end if
11 print p- q + r- 2
```

- 3
- -5
- -12
- -4

Q11. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=2, q=9, r=12
3  if((r&q)<p)
4    r=q+p
5    r=p+r
6  End if
7  Print p+q+r
```

- 18
- 23

26

33

Q12. What will be the output of the following pseudocode?

```
Integer pp,qq,rr
Set pp=5, qq=7, rr=8
for (each rr from 4 to 7)
   if ((pp-qq+rr)<(rr-pp))
      continue
   End if
   qq=pp^qq
   pp=(3+12)+rr
End for
Print pp+qq
```

54

38

35

41

Q13. What will be the output of the following pseudocode?

```
Integer p,q,r
Set p=3, q=10, r=12
p=(p+4)+q
q=q+r
Print p+q+r
```

58

72

68

51

Q14. What will the output of the following pseudocode?

```
Integer p,q,r
Set p=3, q=8, r=4
q=(r+10)+p
p=(4+6)+q
if((8&q)<(p&8))
   p=(p^1)+r
End if
Print p+q+r
```

- 51
- 49
- 53
- 58

Q15. What will be the output of the following pseudocode?

```
1  Set m=1, T=0, n=3
2  if(m>n)
3    m=m-1
4    Go to step 8
5   else
6    T = T + m
7    m = m*T
8   Go to step 2
9  Print m
```

- 12
- 18
- 7
- None of the mentioned options

Q16. The process of copying data from a memory location is called

- Reading
- Writing
- Controlling
- Booting

Q17. When changes occur in a data item, if every file which contains that field should not be updated then, it leads to

- Data Redundancy
- Data Inconsistency
- Data Security
- Data Loss

Q18. To connect a computer with a device in the same room, you might be likely to use

- A coaxial cable
- A dedicated time
- A ground station
- All of the mentioned options

Q19. A distributed network configuration in which all data/information passes through a central computer is

- Bus Network
- Star Network
- Duplex
- Multiplex

Q20. The Boolean expression (A + C') (B' + C') simplifies to

- C'+AB'
- C'(A'+B)
- (BC)'+A'
- None of the mentioned options

Q21. Boolean expression for the output of X-NOR (equivalence) logic gate with inputs A and B is

- AB' + A'B
- (AB)'+ AB
- (A' + B) (A + B')
- (A' + B') (A + B)

Q22. The binary representation 100110 is numerically equivalent to

- The decimal representation 46

- The octal representation 46
- The hexadecimal representation 46
- The binary representation 26

Q23. Which of the following computers are lower than mainframe computers in terms of speed and storage capacity?

- Mainframes
- Mini
- Super
- Hybrid

Q24. IC chips used in computers are made with

- Gold
- Copper
- Silicon
- Silver

Q25. Which of the following type of computer is mostly used for automatic operations?

- analog
- digital
- hybrid
- remote

Q26. The first electromechanical machine “Tabulating machine” was invented by

- Herman Hollerith
- John Napier
- Pascal

Howard Aiken

Q27. In which storage device, recording is done by burning tiny pits on a circular disk

- punched cards
- floppy disk
- magnetic tape
- optical disk

Q28. When we move from the outer most track to the innermost track in a magnetic disk, the density

- increases
- decreases
- remains the same
- either remains constant or decreases

Q29. The current generation of computers

- Second
- Fifth
- Fourth
- Third

Q30. Which of the following is an advantage of the database approach

- elimination of data redundancy
- ability to associate related data
- increased security
- All of the mentioned options

Answer Key & Solution

Section 1 - Aptitude

Q1

392

Solution

We have $49[1 + (R/100)]^4 = 196$

$\Rightarrow [1 + (R/100)]^2 = 2$,

Now $196 [1 + (R/100)]^2 = 196 \times 2 = 392$

Q2

Rs. 16 crores

Solution

Total earnings by Hansie in seven games (in crores) = 10 + 15 + 21 + 12 + 18 + 19 + 17 = 112.∴ Average earning by Hansie in seven games

=Total earnings / Total games= 112/7 = Rs.16 Crores

Q3

9450

Solution

The students must choose 8 from part A and 4 from part B

10c8 x 10c4 = 45 x 210 = 9450

Q4

$16C_2$

Solution

Total number of balls = 9 + 3 + 4

Two balls can be drawn from 16 balls in $^{16}C_2$ ways.

Q5

1/9

Solution

In two throws of dice, n(S) = 6*6 = 36

Let E = event of getting a sum 9 = {(3, 6), (4, 5), (5, 4), (6, 3)}

So, P(E) = n(E)/n(S) = 4/36 = 1/9

Q6

3/11

Solution

Total numbers of balls = 12

No.of.ways to select 3 balls from 12 balls

n(S) = 12C3 = 220

In order for the balls to be different, one ball should be taken from each color

n(E) = 3C1 ×5C1×4C1 = 4 × 3 × 5 = 60

P = 60/220 = 3/11

Q7

40 Days

Solution

Let technicians be M and trainees be W(4M+6W)×8D = 32M+48W

(3M+7W)×10D = 30M+70W

= > 32M-30M = 2M

=> 70W-48W = 22W

2M = 22W

1M = 11W

Hence, (4M+16W) x 8= 50×8 =400

400/10 = 40 DAYS.

Q8

55 days

Solution

We know that work = M × D × H

Here,

M = no. of men

D = no of days

H = hours

So,

$M_1 \times D_1 \times H_1 = M_2 \times D_2 \times H_2$

$11 \times 25 \times 18 = 18 \times D_2 \times 5$

$\therefore D_2 = 55$ days

Q9

54 kmph

Solution

We know that

Speed = Distance/Time

We have the train passing the pole in 15 sec

Hence speed of the train is 150/15 m/sec

= 10 m/sec

Now since the trains cross each other, distance covered = 150 + 150 m

= 300 m

And relative speed = 300/12 = 25 m/sec

We have speed of first train + speed of second train = 25 m/sec

⇒ 10 + speed of second train = 25 m/sec

Speed of second train = 15 m/sec

= 15 × 18/5 kmph

=54 kmph

Q10

2 hrs

Solution

train covered the distance of tunnel ,as well as its length.

So, the total distance travelled by the train is= 58 km + 2 km=60 km

now time taken =60 km / (30 km/hr)= 2 hours

Q11

Rs 10.8

Solution

Let price of 1kg tea be Rs.x and price of 1kg sugar be Rs.y.

x+4y = Rs. 88.20_____(1)

Now, 10% increase in 1kg of Tea = 1+ 10% of 1 = 1 + 0.1 = 1.1 kg of Tea

and 50% increase in 4kg of Sugar = 4+ 50% of 4 = 4+2 = 6 kg of Sugar

Hence, second equation will be:

1.1x + 6y = Rs.114.30_____(2)

equation(1)x11 - equation(2)x10,

y=10.8

Q12

54%

Solution

60% of 90 + 50% of 120 = 54 + 60 = 114 boys out of (120 + 90) = 210.

Required percentage = (114/210) x .100 = 54.28% = 54%.

60% of 90 + 50% of 120 = 54 + 60 = 114

Total = 120 + 90 = 210

Required percentage = 114 / 210 = 0.5428 = 54%

Q13

Rs.404.80

Solution

Let CP of article A = Rs. x

CP of article B = Rs. (600 - x)

SP of article A= Rs. (122x / 100)

SP of article B = Rs. [92(600 - x)/100]

∴ {122x/100} + {92(600 - x)/100} = 600

⇒ 122x - 92x = 60000 - 92 x 600

⇒ 30x = 4800

⇒ x = 160

∴ CP of the article sold at a loss = (600 - x) = Rs 440

Now, we can find the selling price of article B using the loss percentage:

Loss% = 8%

Loss = (8/100) * 440

Selling Price of article B = Cost Price of article B - Loss

Selling Price of article B = 440 - (8/100) * 440

Selling Price of article B = 440 - 35.2

Selling Price of article B = Rs. 404.80

Therefore, the selling price of article B is Rs. 404.80.

Q14 60%

Solution

Let the CP be Rs.100

PROFIT = Rs. 300

SP = 400

New CP = 160% of 100 = Rs. 160

So, the profit will be = 400 - 160 = 240

PROFIT% = 240/400 * 100

=60%

Q15 Rs.160000

Solution

Let the sum be Rs. X. Then, $x(1 + R/100)5 = 2x \Rightarrow (1 + R/100)5 = 2$

The amount after 20 years.

$= x(1 + R/100)20 = x [(1 + R/100)5]4 = 16 \times 10000 = Rs.160000.$

Ans(1)

Q16 20%

Solution

$A= P(1 + R/100)^N$

$20736= 12000(1+ R/100)^3$

$(1+ R/100)^3= 20736/12000= 1.728$

$(1+ R/100)^3 = 1.728$

$1 + R/100= 1.728^{1/3}= 1.2$

$R/100= 0.2$

$R= 20\%$

Q17

179

Solution

New weight = 95 + 7(12)

= 179

Q18

10

Solution

if x no. of question is correct and y no. of question is wrong, then 8x + 5y=0 and x+y=26

by solving we get x=10 & y=16

Q19

6 : 4 : 3

Solution

Let 2x = 3y = 4z = k ,

then x = k/2 , y = k/3 z = k/4

x : y : z = k/2 : k/3 : k/4

1/2 : 1/3 : 1/4

6 : 4 : 3.

Q20

1:1

Solution

The volume of base solution found in the final mixture is the sum of the volumes of base solution in the 30% and 50% solutions.

0.4x + 0.4y = 0.3x + 0.5y

Or 0.1x = 0.1y

x:y = 1:1

Q21

697

Solution

Series Pattern Given Series

582 582

582 + 169 = 751 751

751 – 196 = 555 555

555 + 169 = 724 724

724 – 196 = 528 528

528 + 169 = 697 697

Q22 251

Solution

$43 + 2^2 = 47$

$47 + 3^2 = 56$

$56 + 5^2 = 81$

$81 + 7^2 = 130$

$130 + 11^2 = 251$

Q23 Maternal Grandson

Solution

Male(+), Female(-), Couple(=), Siblings(–), Next Generation (|)

Man
|
Man's son (+) -- Man's son's sister (-)
|
Lady's husband (+) = Lady (-)

From the statement we can clearly understand the son of the man in the photo had a sister, whose son is married to the lady. We also get the information that the man's son and the lady's mother-in-law are siblings, which could only mean that the man is the grandfather of her husband. So, the lady's husband is the maternal grandson of the Man.

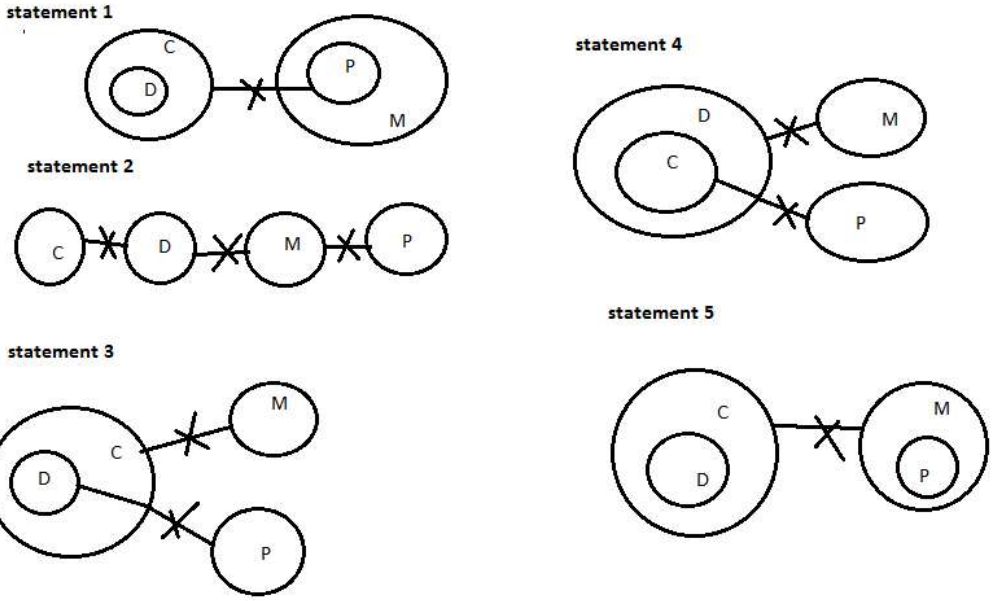
Q24 Uncle

Solution

No Solution

Q25 Only Statements – 1

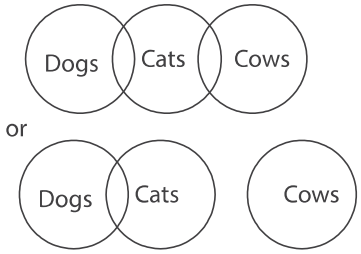
Solution



Only statement 1 follows the conclusion

Q26 if no conclusion follows.

Solution



No conclusion follows. So option 4.

Q27 Either I or II is sufficient

Solution

I. In 'right or wrong' and 'he is right', the common word is 'right' and the common code word is 'nik'. So 'nik' means 'right'. In 'right or wrong' and 'that is wrong', the common word is 'wrong' and the common code word is 'te'. So, 'te' means 'wrong'.

Thus, in 'right or wrong', 'sa' is the code for 'or'. II. In 'that right man' and 'this or that', the common word is 'that' and the common code word is 'pa'. So, 'pa' means 'that'. In 'this or that' and 'tell this there', the common word is 'this' and the common code word is 'ne'. So, 'ne' means 'this'. Thus, in 'this or that', 'sa' is the code for 'or'.

Q28 Both I and II are sufficient

Solution

From I, we conclude that Kapil was born on any one of the dates among 16th, 17th and 18th.

From II, we conclude that Kapil was born on any one of the dates among 13th, 14th, 15th and 16th.

Thus, from both I and II, we conclude that Kapil was born on 16th August.

Q29 if both conclusion 1 and conclusion 2 are true

Solution

Combined Inequalities:

I ≤ S < X = F ≤ K I ≤ S < X = F ≤ K∴ I < K.

Hence, Conclusion I is true.

I ≤ S < X = F ≤ K∴ I < X. H

Hence, Conclusion II is true.

Q30 If both conclusion I and II are true.

Solution

Given statements:

U = V ≤ W > Q ≥ N……(i) B < Q …..(ii) E = W……(iii) From (i) and (iii), we get E = W > Q ≥ N ……(iv) From (iv), E > Q is true.

From (i) and (ii), W > Q ≥ N > B Thus, W > B is true.

Hence, both conclusion I and II are true.

Q31 EDRIRL

Solution

The first, second, third, fourth, fifth and sixth letters of th word are respectively moved two, three, four, five, six and seven steps forward to obtain the corresponding letters of the code.

Q32 22 -21 -25 -26

Solution

Each letter is coded by sum of its position in the alphabetical order and 6.

From the given code, we know that T is 26, so the answer is either options 1 or 4.

The second number should be 1 less than the first number as P comes immediately after O

22 - 21 - 25 - 26

Q33 UAV

Solution

In this series, the third letter is repeated as the first letter of the next segment. The middle letter, A, remains static. The third letters are in alphabetical order, beginning with R.

Q34 pqrst

Solution

The series is mno/nopq/opqrs/pqrst.

Q35 If the data in statement II alone is sufficient to answer the question while the data in statement I alone is not sufficient.

Solution

From I alone:

apple juice **is cedar** == ka **la do** ni

cedar vinegar **is** good == **do** re **la** mi

buy some juicy apples == vu hi wo pe

So, we get:

Is/cedar == la/do (not necessarily in the same order)

Apple/juice ==ka/ni (not necessarily in the same order)

Vinegar/good == re/mi (not necessarily in the same order)

So, the code for 'juicy apple' cannot be determined.

So, statement I alone is not sufficient.

From II alone:

good *orange and* **apple** == **ni** *xe co* mi

apple juice is **fresh** == ka la **ni** **po**

fresh *and* juicy *orange* == xe co pe **po**

so, we get

Apple == ni

And/ orange == xe/co (not necessarily in the same order)

Fresh == po

so, juicy == pe

So, the code for 'juicy apple' is 'ni pe'.

So, statement II alone is sufficient.

Q36

If the data in statement II alone is sufficient to answer the question while the data in statement I alone is not sufficient.

Solution

From I alone : we have one final arrangement :

There is no information about point Q.

Hence, I alone is not sufficient.

From II alone: we have 2 different arrangement :

From point O, he moves towards north and takes a left turn to reach point Q, when he reached point Q, he is facing WEST (since he took left turn while walking towards north)

Hence, II alone is sufficient.

So, the answer is II ALONE IS SUFFICIENT

Q37

25P

Solution

- 2D: Starts with the number 2 followed by the letter D.

- 6E: Starts with the number 6 followed by the letter E.
- 14J: Starts with the number 14 followed by the letter J.
- 25P: Starts with the number 25 followed by the letter P.

The pattern seems to be the number increasing sequentially, but the letter does not follow any obvious pattern.

The odd one out is 25P because it does not follow the same pattern as the others. The number is higher than expected, and the letter is also different from the pattern established by the previous elements.

So, the correct answer is: 25P

Q38 CFG

Solution

If the conclusion might not stand up, it suggests that the conclusion was a bad one. If we select 'hasty' for this second blank, then it is easy to see that the thinking was done 'too quickly' and that 'slower' scrutiny would reveal the error. (Precipitous – hurried; warily – cautiously)

Q39 block

Solution

The word 'block' fills the blanks in A ('block letters') means 'capital letters', B ('block' means 'a large building' especially when subdivided into flats), C ('put the head on the block' mean 'risking reputation of position by taking particular course of action') and D ('block' means to obstruct a passage etc.). Words 'capital' (a), stop (c) and 'range' (d) fill no blank; options (1), (c), (d) are wrong.

Q40 throughout her school and college years but lately got

Solution

The sentence is not about the present, so options 1 and 4 can be eliminated.

Her singing, dancing and acting is not the same as it was in school and college, so throughout is better than since.

Q41 was taken aback

Solution

"Was taken aback "is the correct answer. The statement represents an action that happened in past. Hence option "will be taken aback" is eliminated as it refers to a future action. 'Taken away' is incorrect as he was not taken away by anyone. Hence options had been taken away and would be taken away are ruled out. **'Taken aback' means 'confused' or 'shaken'**. It is the best fit for this context.

Q42 has always been and will continue to be feared until socialism takes over

Solution

The given sentence is correct except for the last part 'socialism will have taken over.' This should be in the simple present tense.

Q43

Does he know

Solution

The word does is in simple present tense, and so thought would not be correct. Part c is equally confusing but the question mark helps us to eliminate the option. Having a sentence like " Will he have to face the consequences.The answer is "does he know"

Q44

Antagonist

Solution

The word that best describes Rakesh in this situation is antagonist.

An antagonist is a character in a story who opposes the protagonist. In this case, Rama is the protagonist, as he is the one who is being praised for his presentation. Rakesh, on the other hand, is opposing Rama, so he is the antagonist.

Q45

Elevate

Solution

Subvert means undermining the power and authority of an established system or institution. The opposite of this should be uplifting. So from the options elevate is the Antonym.

Q46

Temporary suspension

Solution

Abeyance (Phrase) : If something is **in abeyance**, it is not operating or being used at the present time or for a period of time.

Q47

The villagers took Rohit to the hospital

Solution

The given statement is in Passive, hence it needs to be converted to Active.

The villagers becomes subject, Rohit becomes object.

was taken (be + V3 form of take) becomes 'took' (V2 from of take)

Hence, The villagers took Rohit to the hospital.

Q48 They will have vacated the participants by now.

Solution

By now the participants will have been vacated

They have vacated the participants by now. (tense changed)

They will have been vacated the participants by now. (Tense has changed)

They will have vacated the participants by now ---> is correct

Hence, They will vacate the participants now. (Tense has changed)

Q49 BDAC

Solution

Ans. (2). 1B, BD and C6 are the keys. After the introductory statement, 'At the heart of the novel...' introduced by 1, B should follow 1 by stating 'in his ramshackle boat...'. D, A and C further describes the novel. D must follow B by telling 'The river unites them...', as 'boat' in B is related to 'river' in D. A must succeed D with 'He can read the secrets...'. C should follow A with 'They have no words...'. 6 should follow C as it provides an appropriate conclusion for the passage by stating 'Silence and gestures, movements of eye 'and figures...'. So, BDAC is the right sequence, Hence, option (2) is the correct option.

Q50. 42

Solution

Required average = (47 + 55 + 24)/3 = 126/3 = 42 lakhs

Q51. 75

Solution

Investment made by A and E in March = 52 + 83 = 135

Investment made by B and D in January = 28 + 32 = 60

Hence, required difference = 135 – 60 = 75

Q52. 23%

Solution

Required percent = (80 – 62)/ 80 × 100 = 18 / 80 × 100 = 22.5% = 23%

Q53. Might

Solution

ı the first paragraph , the line "Thus in disputes between man and man ..... taken the lace of might." means that in the older days it was might that was rule of the land.

Q54. Better health

Solution

First two lines of the second paragraph has the answer in it.

"Better health" is the right option

Q55. Order and safety

Solution

The first line of the paragraph talks about the first merit of civilization which is Order and Safety.

Q56. The present civilization is more secure than any that has gone before

Solution

In the second paragraph, the last before line as well as the last line have the answer in it.

"The present civilization is more secure than any that has gone before" is the right option.

Q57. Freedom from the fear of pain

Solution

The right option is "Freedom from the fear of pain". This can be inferred from the second paragraph

Q58. Seamless

Solution

seamless-smooth and continuous, with no apparent gaps or spaces between one part and the next.

The lines before the blank talk on the benefit of cloud computing and how effortlessly and continuously files are transferred.

Seamless is the right answer

Q59. Lightening

Solution

ghtening fast is a phrase meaning very fast

Q60. Viable

Solution

viable- feasible

The line talks on how difficult data transfer was before cloud computing

Section 2 - Pseudocode and Computer Fundamentals

Q1 It will print Hello one time

Solution

No Solution

Q2 33

Solution

Here q=3+6=>9

p=9&4=>0

r=9+6=>15

p=0+9=>9

So 9+9+15=>33

Q3 39 32

Solution

Initially b=7,c=1

When j=2,i=1,then c= 2

When j=2,i=2,then c= 4

When j=3,i=1,then c= 8

When j=3,i=2,then c= 16

When j=3,i=3,then c= 32

Finally loop terminates and print b=7+32=>39 and c=32. So 39 32 gets printed.

Q4 19

Solution

Value of 'b' set to 14 after execution of for loop.

So, 5 + 14 = **19**

Q5

15

Solution

Initially a=0, b=6, c=9, for all the loop iteration if block gets failed and finally a=9, b=6 so 15 gets printed.

Q6

3

Solution

No Solution

Q7

42

Solution

No Solution

Q8

12

Solution

Initially a =5 , b=300 and c=1

Inside loop:

while b=0 : a incremented by 1 (a=6)

while b=1 : a incremented by 1 (a=7)

while b=2 : a incremented by 1 (a=8)

while b=3 : a incremented by 1 (a=9)

while b=4 : a incremented by 1 (a=10)

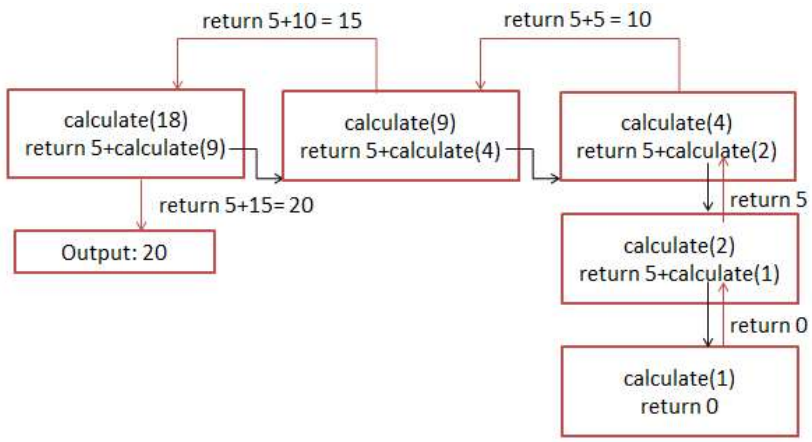
while b=5 : a incremented by 1 (a=11)

Then 'a' is added with 'c' (11+1) and prints 12.

Q9

20

Solution



Function call starts from calculate(18), then it calls itself until 'b' reaches 1. Function returns 0 to calculate(1) ,calculate(2) return 5 by adding 5+0. Same process repeated upto calculate(18). In calculate(18), 5+15 = 20 . Final value(20) returned to main method and gets printed.

Q10

-5

Solution

Initially p=1, q=4, r=2

1^4 = 5 , 1&4 =0 , so if block executes r=1&4 =0

Second if block gets failed and print 1-4+0-2 = **-5**

Q11

23

Solution

If condition gets failed so 2+9+12=>23 gets printed.

Q12

38

Solution

When rr=4,updated value is pp=19and qq= 2

When rr=5,updated value is pp=20 and qq= 17

When rr=6,updated value is pp=21 and qq= 5

When rr=7,updated value is pp=22 and qq= 16

Finally 22+16=38

Q13

51

Solution

No Solution

Q14

51

Solution

No Solution

Q15

None of the mentioned options

Solution

No Solution

Q16 Reading

Solution

No Solution

Q17 Data Inconsistency

Solution

No Solution

Q18 A coaxial cable

Solution

No Solution

Q19 Star Network

Solution

No Solution

Q20 C'+AB'

Solution

No Solution

Q21 (A' + B) (A + B')

Solution

No Solution

Q22 The octal representation 46

Solution

No Solution

Q23 Mini

Solution

No Solution

Q24 Silicon

Solution

No Solution

Q25 hybrid

Solution

No Solution

Q26 Herman Hollerith

Solution

No Solution

Q27 optical disk

Solution

No Solution

Q28 increases

Solution

No Solution

Q29 Fifth

Solution

No Solution

Q30 All of the mentioned options

Solution

No Solution